Differential Neural Activity During Decision Making Reflects Both Reward Impulsivity and Decision Difficulty

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Introduction

- Decision making impulsivity (DM-impulsivity) is reflected by short reaction times (RT) in choice tasks and involves fast responses executed by quick neural processes. [1]
- Reward-impulsivity (R-impulsivity) is observed as a preference for smaller rewards sooner (SS) over larger rewards later (LL) as assessed by delay discounting (DD) tasks. [2]
- All DD choices are postulated to reflect information integration of reward amount and time. R-impulsive subjects minimize the time to reward while R-patient subjects maximize the reward amount.

Hypotheses

- DD choices include both R-impulsivity and DM-impulsivity. R-impulsive subjects will be more likely to make SS choices fast, whereas R-patient subjects will be more likely to make LL choices fast.
- Slow DD choices reflect reward amount and time integration while fast DD choices reflect DM-impulsivity, therefore we expect to see the DD task engage different brain processes as a function of fast and slow RT, irrespective of whether subjects are R-impulsive or R-patient.

Analysis

- Behavioral: A Bayesian generalized linear mixed-effects model was performed on SS vs. LL choices as a function of reward Trial Type, Framing, Reward Time Difference, SS Reward Amount, Relative Reward Difference between SS and LL reward amount (RRD), RT (continuous), Group and their interactions.
- Imaging: a) Standard preprocessing (slice timing and motion corrections, normalization and smoothing) and individual and group analyses were performed using SPM12. Individual-level analysis investigated the effect of RT Type (3 levels) using regressors for short, medium and long RTs with RRD as a parametric modulator. b) RT of all subjects were classified as short (400-1500ms), medium (1500-3500ms) and long RT (> 3500ms). We divided the participant to fast and slow groups based on their RT distribution. c) Individual subjects' results were submitted to random-effects whole brain and ROI group analyses. One-sample t-tests were used to identify differences between RT types (within and between participants), using ROI analyses. d) ROI analysis included brain regions that appear active in the majority of DD tasks [3].

Behavioral Results

- As expected, the likelihood of choosing SS was significantly affected by SS amount, Reward Time Differences, and RRD.
- The interaction of RT X Group (Fig. 2A) demonstrates that the R-impulsive group was more likely to make faster SS choices and that the R-patient group was more likely to make faster LL choices.
- The interaction of RT X RRD (Fig. 2B) shows that only small RRDs eliciting longer RTs were associated with LL choices. RTs were not associated with choices for large RRDs.

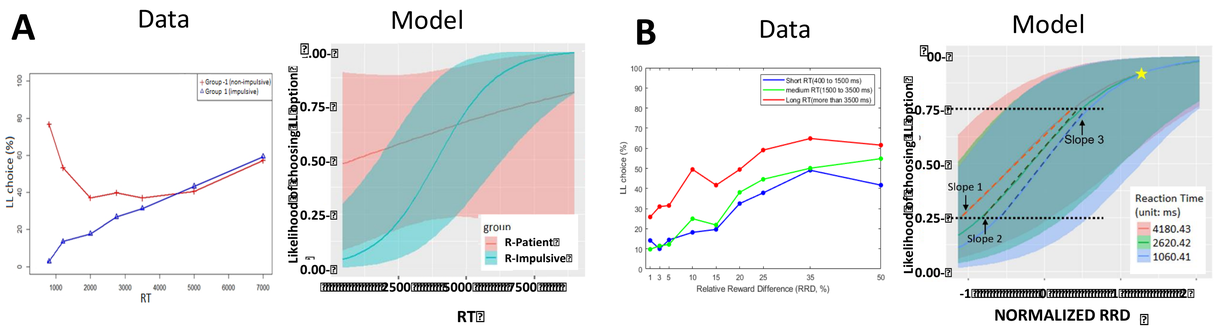


Figure 2. Data and Model predictions of participant's choice preferences. A. DD choices as a function of RT for each R-impulsivity group. B. DD choices as a function of RRD with varied RT responses. For illustration purposes, we included the distribution of the data as a function of the 3 RT types.

Study Design

- Participants (N=20; age 22.5±4.0; female=7) performed a DD task (Fig.1) (on average 124 trials) in an MRI scanner.
- The participants were divided into R-impulsive (N=10; 4F) and R-patient (N=10; 3F) groups based on previous performance on the same task.
- The DM-impulsive (Fast) (N=10) and DM-patient (Slow) (N=10) participants were evenly distributed across the R-impulsivity groups.

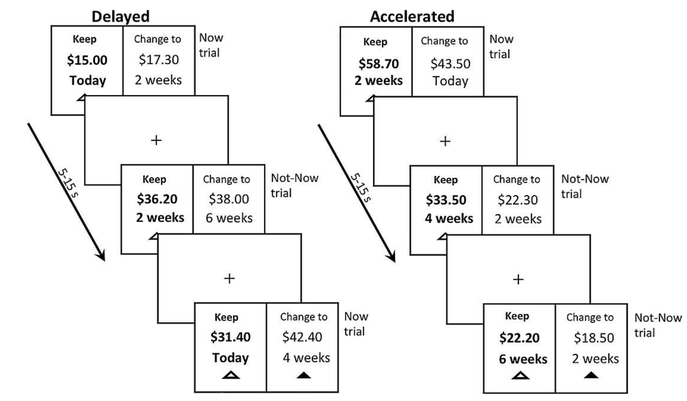
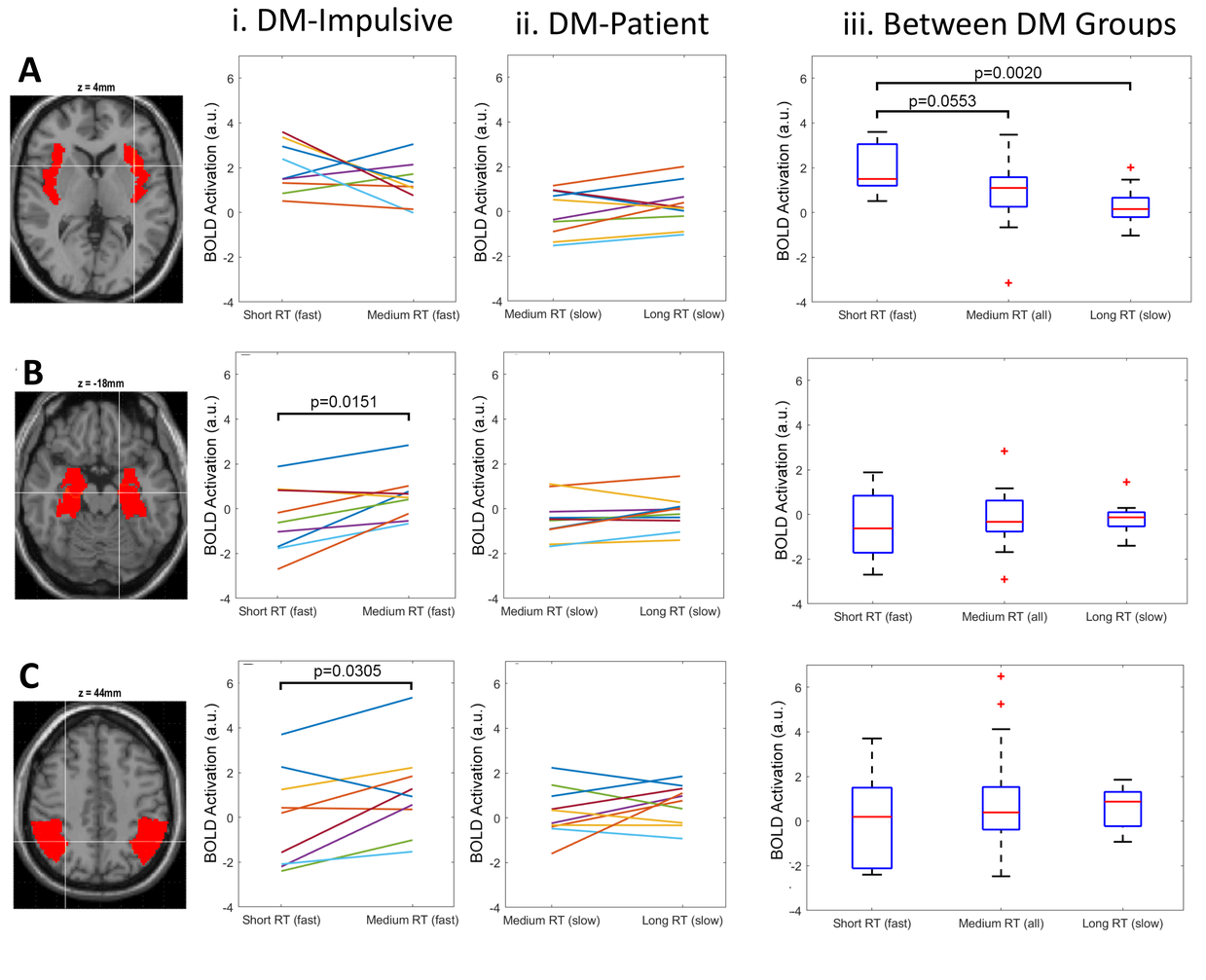


Figure 1: DD task design showing SS and LL amounts & times, trial type (Now vs. Not-Now), and framing (delayed vs. accelerated) conditions.

Brain Imaging Results

- ROI analysis revealed significant differences in BOLD signal as a function of RT Types and DM-impulsivity groups.
- Insula BOLD activity of the DM-impulsive group, who have short RTs, was stronger than that of the DM-patient group, who responded slowly with long RT. (Fig. 3Aiii). No differences within each DM-impulsivity group were observed when comparing RTs. (Fig. 3Ai and 3Aii)
- Stronger BOLD activity in the Parahippocampus and Inferior Parietal cortex was found among DM-impulsive group when comparing short and medium RTs. (Fig 3Bi and 3Ci). No differences within the DM-patient group or between groups were found.

Figure 3. Comparison of BOLD activity within DM-impulsive and DM-patient groups, and between DM groups for A. Insula, B. Parahippocampus, and C. Inferior Parietal



Conclusion

- ✓ The behavioral data confirm our first hypothesis, demonstrating that when subjects made their choices fast (DM–impulsivity), they could be either R-impulsive choosing SS, or R-patient choosing LL.
- ✓ The MRI data supports our second hypothesis, demonstrating differential brain mechanisms for fast and slow choices both within and between DM-impulsivity groups.
- ✓ Both SS and LL fast responses made by the DM-impulsive group differed than choices made by the DM-patient group, suggesting differences in decision strategies. The large Insula activity seen in the DM-impulsive group suggests habitual action selection [4], or an *a priori* decision, which can occur fast, whereas decisions made by the DM-patient group were more deliberate.
- ✓ Slow responses (medium RTs) made by the DM-impulsive group suggest that DD choices were more difficult and therefore involved a more complex brain activation demonstrated by increased BOLD signal in the Inferior Parietal cortex and the Hippocampus. The role of the Inferior Parietal cortex in cognitive and executive control [5] and of the Hippocampus in anticipation of reward [6] support the idea that slow DD choices engage integration of amount and time information.
- ✓ The DM-patient group did not elicit differential activity as a function of RT for any ROI, suggesting that they engage similar brain processes for both fast and slow DD choices.

Acknowledgments

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References

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